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Continuous Integration with CodeBuild



yakubu abdullahi yusuf

```
1 version: 0.2
2
3 phases:
4   install:
5     runtime-versions:
6       java: corretto
7   pre_build:
8     commands:
9       - echo Logging in to AWS CodeArtifact...
10      - CODEARTIFACT_AUTH_TOKEN=$(aws codeartifact get-authorization-token --domain nextwork --domain-owner $your_aws_account_id --)
11      - export CODEARTIFACT_AUTH_TOKEN
12   build:
13     commands:
14       - echo Build started on `date`
15       - mvn clean install -s settings.xml
16   post_build:
17     commands:
18       - echo Build completed on `date`
19       - echo Packaging artifacts...
20       - mvn package -s settings.xml
21   artifacts:
22     files:
23       - target/*.war
24     discard-path: no
```



yakubu abdullahi yusuf

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Introducing Today's Project!

Day 4 of the CICD 7 days DevOps challenge. In this project, I will create an d configure CodeBuild project, connect the CodeBuild to GitHub repository, define my build using spec.yaml and automate testing using CodeBuild, stay tuned and enjoy.....

Key tools and concepts

Services I used were EC2 instance, S3 buckets, CodeBuild, IAM policies and Roles
Key concepts I learnt include configuring build pipelines with CodeBuild, managing source code builds using buildspec.yml, integrating GitHub with AWS services, and understanding how cloud resources work together to automate application deployment.

Project reflection

This project took me approximately 50 minutes and the most challenging part was the error i got in the build limit, It was most rewarding to see my build running successfully.

This project is part four of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project.



yakubu abdullahi yusuf

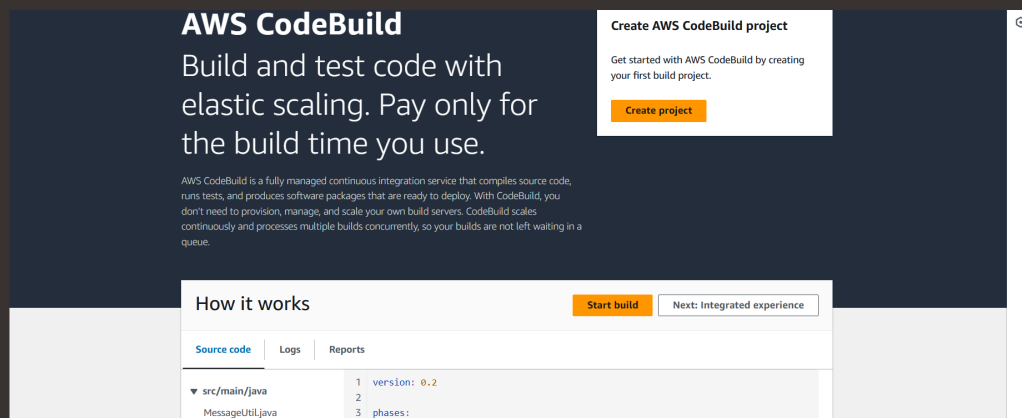
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Setting up a CodeBuild Project

CodeBuild is a continuous integration service, which means it automatically builds, tests, and packages your code every time you make changes. Engineering teams use it because it doesn't require manual set up or managing servers manually.

My CodeBuild project's source configuration means it tells CodeBuild what to build and where to get it from and I selected GitHub as the Source provider.





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Connecting CodeBuild with GitHub

There are multiple credential types for GitHub, like token, Oauth and Github App. I used GitHub App because it is the simplest and it is the most secure. It eliminates the need to work with tokens or other separate applications.

The service that helped connect AWS environment to GitHub is AWS CodeConnections.

Source 1 - Primary

Source provider
GitHub ▼

Credential
✔ Your account is successfully connected by using an AWS managed GitHub App. [Manage account credentials.](#)
☐ Use override credentials for this project only

Repository
☒ Repository in my GitHub account
☐ Public repository
☐ GitHub scoped webhook

Repository



CodeBuild Configurations

Environment

My CodeBuild project's Environment configuration tells AWS what kind of resources my build should run on and what kind of compute environment my build will use. it includes settings like compute type, environment type, provisioning model etc.

Artifacts

Build artifacts are the outputs of my build processes. They're important because they are what i will deploy to my servers or users. My build process will create the build artifacts so to store them, I created an S3 bucket.

Packaging

When setting up CodeBuild, I also chose to package artifacts in a zip because it makes it smaller in size, easily maintained and it will be organized in a proper way.

Monitoring

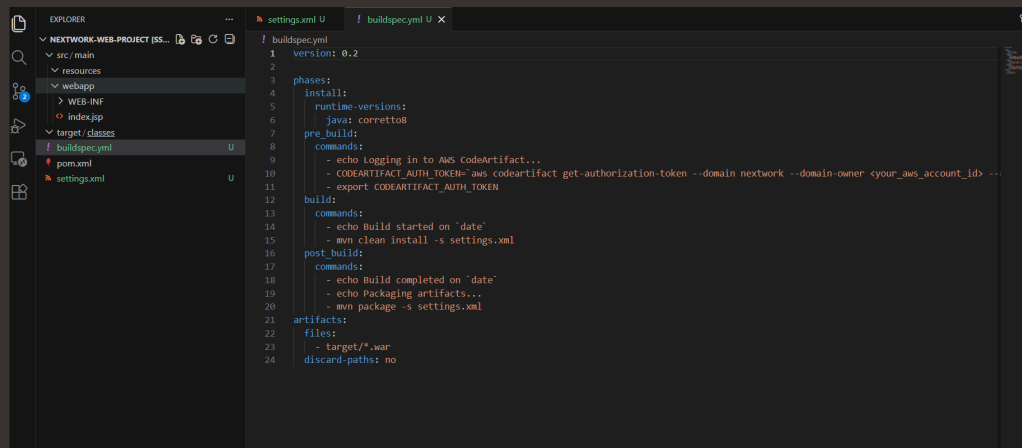
For monitoring, I enabled CloudWatch Logs, which is a monitoring service that collects and tracks logs from AWS services. It records everything that happens during build processes. It is very helpful in terms of debuggings.



buildspec.yml

My first build failed because a build spec file was missing. A buildspec.yml file is needed because it serves as an instructions manual that tells our build what step it should follow. Without it, the build has no guidance on what steps to execute.

The first two phases in my buildspec.yml file are install and pre_build which tells the codebuild to use Java8 and grabs security tokens for a secure access of dependencies respectively . The third phase in my buildspec.yml file is build, the build phase is where the building takes place, it also compiles the code. The fourth phase in my buildspec.yml file is postbuild, which is the final touches of the build and packages everything into a WAR file.



```
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Success!

My second build also failed, but with a different error that said CodeBuild service role doesn't have permission to access CodeArtifact. To fix this we need to grant CodeBuild's IAM role the permission to access CodeArtifact.

To resolve the second error, I added a policy to my code build service role. When I built my project again, It built successfully.

To verify the build, I checked my S3 bucket, seeing the artifact tells me that the build was successful.

Build status

Status
✔ Succeeded

Initiator
ABDOU-IAM-Admin

Build ARN
 arn:aws:codebuild:us-east-2:176971016201:build/nextwork-devops-cicd-rabi:93d71dd9-13d6-4047-bafa-68b3de4acb71

Resolved source version
ec021b09b7df5b17bda9875c9d676a0bbf465eda

Start time
Jan 23, 2026 1:15 AM (UTC+3:00)

End time
Jan 23, 2026 1:16 AM (UTC+3:00)

Build number
5

Build logs

Phase details

Reports

Environment variables

Build details

Resource utilization

Name	Status	Context	Duration	Start time	End time
SUBMITTED	✔ Succeeded	-	<1 sec	Jan 23, 2026 1:15 AM (UTC+3:00)	Jan 23, 2026 1:15 AM (UTC+3:00)
QUEUED	✔ Succeeded	-	<1 sec	Jan 23, 2026 1:15 AM (UTC+3:00)	Jan 23, 2026 1:15 AM (UTC+3:00)



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